Integrated Spoof-Aware Speaker Verification (SASV)

Comprehensive Project Guide: Non-Technical Intuition, Mathematical Foundations, Deep Learning Architecture, and Technical Interview Playbook

Target Domain: Biometric Voice Security	**ASV Model:** ECAPA-TDNN (192-d)	**CM Model:** Raw-wave AASIST
Fusion Layer: Trained LogReg + CM Gate	**Liveness:** Whisper ASR Dynamic Challenge	**Hardware:** PyTorch CUDA (RTX 3050)

PART 1: NON-TECHNICAL GUIDE (UNDERSTANDING THE SYSTEM)

1.1 Executive Summary & Real-World Motivation

In modern consumer and enterprise technology, voice is rapidly becoming the most convenient password. Banking apps, customer call centers, automotive assistants, and smart home locks allow users to authenticate themselves simply by speaking. However, the generative AI revolution has shattered the foundational assumption of traditional voice authentication: **that only the real person can sound like the real person**.

Using modern deep learning tools (such as ElevenLabs, VALL-E, or open-source neural vocoders), an attacker can capture just 3 to 10 seconds of someone's voice from social media or a phone call and synthesize an uncanny, photo-realistic voice clone. Furthermore, physical replay attacks—where someone records your voice on a high-end smartphone and plays it back into the microphone—are effortless to execute. This project implements an **Integrated Spoof-Aware Speaker Verification (SASV)** system that eliminates this blindspot.

Mental Model: The Bank Vault Guard

The Bank Vault Analogy:
Imagine walking up to a high-security bank vault with a physical security guard:
• **Traditional Voice Authentication (ASV):** The guard looks only at your photo ID badge. If an attacker shows up wearing an ultra-realistic 3D latex mask sculpted to look exactly like you, the guard says: *'The face on the badge matches the face standing here. Access Granted!'* Traditional systems fail 100% of the time against voice clones.
• **Our Spoof-Aware System (SASV):** The guard does two distinct checks simultaneously. First, they verify your ID (Voiceprint). Second, they shine infrared light and check your pulse to see if you are a living human breathing in front of them, or a latex mask (Anti-Spoofing). Access is granted only when **both** identity and human authenticity pass.

1.2 The Two-Question Defense Philosophy

Whenever someone speaks into our system to log in, the backend evaluates two distinct, independent questions:

Question 1 (Identity): 'Does this voice sound like the registered owner?'
Handled by the *Automatic Speaker Verification (ASV)* engine. It measures vocal tract resonance, pitch cadence, and acoustic timbre against the enrolled user template.

Question 2 (Authenticity): 'Is this voice a living human, or an AI clone / loudspeaker playback?'
Handled by the *Countermeasure (CM)* engine. It searches for microscopic synthetic artifacts: phase distortions, sub-harmonic electronic buzz, and loudspeaker enclosure reverberation.

Question 3 (Liveness): 'Is this person speaking right now in real time?'
Handled by the *Dynamic Challenge Engine*. The system presents a freshly randomized phrase (e.g. *'green cloud holds near 44'*) with a 20-second countdown. An attacker cannot use a prerecorded audio clip because they cannot predict the phrase in advance.

1.3 Step-by-Step User Journey

Step	What the User Does	What Happens Behind the Scenes	Security Safeguard Enforced
1. Enrollment	User speaks into mic for 5 seconds (e.g. 'Hello, my name is Satheesh...')	Audio is checked for spoofing. If authentic, an AI voiceprint is created and saved.	Attackers cannot enroll with cloned audio; spoof gate blocks synthetic clips.
2. Challenge Request	User clicks 'Authenticate' and receives a random secret phrase.	Server issues a single-use session ID and phrase with a 20-second timer.	Tokens expire after 20s and cannot be reused, blocking replay recordings.
3. Speaking Phrase	User reads the phrase aloud into the microphone.	Whisper AI transcribes speech; verifies both words and the 2-digit number.	Ensures the speaker is physically responsive in real time.
4. Joint Decision	User receives instant access decision (Granted or Denied).	Voice similarity score and anti-spoof score are evaluated jointly with safeguard gate.	Even if a voice clone has 99% similarity, the CM gate blocks it as SUSPECTED_SPOOF.

PART 2: TECHNICAL ARCHITECTURE & MATHEMATICAL FOUNDATIONS

2.1 Mathematical Problem Formulation & Trial Taxonomy

In classical Automatic Speaker Verification (ASV), the task is a binary hypothesis test between genuine target speaker and zero-effort human impostor. In **Spoof-Aware Speaker Verification (SASV)**, the hypothesis space expands into a tri-factor taxonomy. Every authentication trial consists of a claimed identity C and an audio probe X:

Trial Category	Speaker Relation	Audio Authenticity	Ground Truth Label	System Action
Target Genuine	True Account Owner	Bona Fide Human Voice	Positive ($y = 1$)	**ACCEPT** (ACCESS_GRANTED)
Human Impostor	Zero-Effort Impostor	Bona Fide Human Voice	Negative ($y = 0$)	**REJECT** (IMPOSTOR_HUMAN)
Target Spoof	Synthesized / Replayed Target	Spoof (Deepfake / Replay)	Negative ($y = 0$)	**REJECT** (SUSPECTED_SPOOF)
Non-Target Spoof	Other Voice Synthesized	Spoof	Negative ($y = 0$)	**REJECT** (DUAL_CRITERIA_FAILURE)

2.2 Official Datasets & Benchmarks Used in This Project

Building a production-grade SASV system requires distinct acoustic corpora for speaker verification pretraining, countermeasure graph learning, challenge evaluation, and live microphone calibration:

Dataset	Scale & Utterances	Acoustic Nature & Labels	Role in Our System
VoxCeleb 1 & 2	1.2M+ utterances, 7,000+ speakers	Wild YouTube interviews, diverse acoustic noise	Pretraining ECAPA-TDNN speaker embedding backbone.
ASVspoof 2019 LA	121,461 trials across Train, Dev, Eval	Logical Access: 19 speech synthesis & voice conversion algorithms (A01-A19)	Pretraining AASIST spectro-temporal graph attention network.
SASV 2022 Challenge Benchmark	102,579 Official Eval Trials	Official protocol matching VoxCeleb speakers with ASVspoof attacks	Formal architectural benchmarking against 6 international baselines.
Live Pilot Calibration Partition	42 trials (21 DEV, 21 EVAL)	Multi-speaker recordings with genuine, impostor, and target attacks	Score fusion weight training (DEV) and gate validation feedback (EVAL).

2.3 Deep Learning Model Architectures

A. Speaker Verification Backbone: ECAPA-TDNN (SpeechBrain)

The ECAPA-TDNN (Emphasized Channel Attention, Propagation, and Aggregation Time-Delay Neural Network) is the state-of-the-art architecture for speaker verification. It processes 80-dimensional log Mel-filterbank acoustic features with three key innovations:

• **1D Dilated Convolutions:** Large receptive field capturing phonetic co-articulation across time.
• **Squeeze-and-Excitation (SE) Blocks:** Channel-attention modules that adaptively rescale feature channels by modeling inter-channel dependencies.
• **Multi-layer Feature Aggregation (MFA):** Concatenates intermediate feature maps from all prior Res2Net blocks.
• **Attentive Statistical Pooling (ASP):** Computes weighted mean and standard deviation vectors across time frames.
• **Cosine Scoring:** Extracts a 192-dimensional L2-normalized embedding vector $\mathbf{e}$. Similarity between test embedding $\mathbf{e}_{\text{test}}$ and enrolled template $\mathbf{e}_{\text{enroll}}$ is computed via dot product: $s_{\text{asv}} = \mathbf{e}_{\text{test}} \cdot \mathbf{e}_{\text{enroll}}$.

B. Anti-Spoofing Countermeasure: AASIST (Raw Waveform Graph Attention)

AASIST (Audio Anti-Spoofing using Integrated Spectro-Temporal Graph Attention Networks) processes the raw 16 kHz audio waveform directly without computing Short-Time Fourier Transforms (STFT), preserving subtle phase discrepancies that neural vocoders distort:

• **SincNet Filterbank:** First layer consists of parameterized band-pass filters learning cutoff frequencies directly from raw data.
• **Spectro-Temporal Graph Attention:** Construct two sub-graphs: a temporal graph (connecting time frames) and a spectral graph (connecting frequency bands).
• **Heterogeneous Graph Attention Network (GAT):** Merges spectral and temporal nodes into an integrated graph, passing attention messages across domains.
• **Readout & Scoring:** A graph pooling layer outputs a 2-dimensional logit vector: bona fide logit l_0 and spoof logit l_1. The countermeasure score is the log-likelihood ratio: $s_{\text{cm}} = l_0 - l_1$.

C. Dynamic Challenge & Liveness Engine: OpenAI Whisper ASR

To defeat physical replay attacks (loudspeaker playback of authentic recordings), the system integrates an automatic speech recognition (ASR) challenge-response module running Whisper 'tiny' directly on CUDA GPU:

• Issues randomized English phrases composed of phonetic tokens plus a 2-digit number (e.g. *'warm sun shows up 81'*).
• Generates a cryptographically random 32-character session ID with a 20-second TTL (Time-To-Live).
• Whisper transcribes the spoken probe; phonetic fuzzy matching computes word overlap (threshold: $\ge 60\%$) and explicitly validates the 2-digit security number.
• Immediately marks the session token as consumed to prevent parallel race-condition replays.

2.4 Fusion Theory & The Mandatory Safeguard Gate

A core contribution of this project is proving both mathematically and experimentally why pure linear score fusion fails against voice clones, and how a dual-criteria safeguard gate eliminates the vulnerability.

A. Linear Score Fusion Model

The system trains a logistic regression fusion layer using PyTorch BCEWithLogitsLoss on DEV scores to map the 2D score tuple $(s_{\text{asv}}, s_{\text{cm}})$ into a single scalar decision score:
$$s_{\text{fus}} = w_{\text{asv}} s_{\text{asv}} + w_{\text{cm}} s_{\text{cm}} + b$$
On our calibrated DEV distribution, the trained weights are: $w_{\text{asv}} = +12.7265$, $w_{\text{cm}} = +0.1767$, $b = -3.4780$.

Critical Vulnerability: Voice Clone Leak in Pure Fusion

The Voice Clone Mathematical Failure in Pure Linear Fusion:
Consider trial EVAL-B-TSPOOF-SYN (a high-quality voice clone attack impersonating enrolled user Beta):
• Speaker Similarity: $s_{\text{asv}} = +0.7181$ (Extremely high identity match)
• Countermeasure Score: $s_{\text{cm}} = -6.0606$ (AASIST flagged synthetic acoustic artifacts)
Now calculate the linear fused score:
$$s_{\text{fus}} = (12.7265 \times 0.7181) + (0.1767 \times -6.0606) - 3.4780 = +9.1389 - 1.0709 - 3.4780 = \mathbf{+4.5899}$$
Because w_{asv} is so large, the high speaker similarity completely overwhelms the countermeasure warning. Under pure linear fusion, since $+4.5899 \ge 0.5052$, **the clone attack is granted full access!**

B. The Mandatory Countermeasure Safeguard Gate

To defeat this vulnerability, our strict_security operational policy enforces a **non-linear tandem safeguard gate**. A candidate probe must satisfy both the global fused score threshold AND a hard countermeasure gate floor:
$$\text{Decision} = \begin{cases} \text{ACCESS_GRANTED} & \text{if } s_{\text{fus}} \ge \tau_{\text{fus}} \text{ AND } s_{\text{cm}} \ge \tau_{\text{cm}} \\ \text{SUSPECTED_SPOOF} & \text{if } s_{\text{fus}} \ge \tau_{\text{fus}} \text{ AND } s_{\text{cm}} < \tau_{\text{cm}} \\ \text{ACCESS_DENIED} & \text{otherwise} \end{cases}$$
When the Beta clone attack is submitted under strict_security ($\tau_{\text{cm}} = -4.0694$), even though $s_{\text{fus}} = +4.5899$, because $s_{\text{cm}} = -6.0606 < -4.0694$, the safeguard gate immediately triggers and returns **SUSPECTED_SPOOF** with 100% attack rejection.

2.5 Reproducible DEV-Only Threshold Selection Procedures

To prevent statistical data leakage, operational policy thresholds are derived using exact arithmetic formulas from the DEV partition distributions alone, without referencing the EVAL partition:

Policy Tier	Exact Derivation Formula	Derived Threshold Values	CM Gate	Operational Use Case
Balanced	Exact linear midpoint boundary between DEV impostor ceiling and genuine floor: $\tau_{\text{fus}} = (\max S_{\text{imp}} + \min S_{\text{gen}}) / 2$ $\tau_{\text{asv}} = (\max S_{\text{imp}}^{\text{asv}} + \min S_{\text{gen}}^{\text{asv}}) / 2$	$\tau_{\text{fus}} = +0.0783$ $\tau_{\text{asv}} = +0.3130$ $\tau_{\text{cm}} = -4.0694$	False	Balanced operational trade-off separating human impostors from legitimate users.

High Convenience	Set directly at the highest observed DEV human impostor score: $\tau_{\text{fus}} = \max S_{\text{imp, dev}}$ $\tau_{\text{asv}} = \max S_{\text{imp, dev}}^{\text{asv}}$	$\tau_{\text{fus}} = -0.3486$ $\tau_{\text{asv}} = +0.3024$ $\tau_{\text{cm}} = -6.5000$	False	Maximizes convenience (100% TAR on DEV). Admitted in low-risk scenarios.
Strict Security	Set directly at the lowest observed genuine DEV scores (DEV floor) with mandatory CM gate: $\tau_{\text{fus}} = \min S_{\text{gen, dev}}$ $\tau_{\text{cm}} = \min S_{\text{gen, dev}}^{\text{cm}}$	$\tau_{\text{fus}} = +0.5052$ $\tau_{\text{asv}} = +0.3237$ $\tau_{\text{cm}} = -4.0694$	**True** (Mandatory)	Bank-grade security: Defeats target clones and replays; rejects 4/4 attacks on EVAL.

PART 3: EMPIRICAL RESEARCH FINDINGS & PROTOCOL RIGOR

3.1 The Beta Countermeasure Score Inversion Phenomenon

A critical empirical discovery emerged during gate evaluation on speaker Beta:

Key Finding: The Beta Score Inversion

The Score Inversion:
• Authentic User Beta (EVAL-B-GEN): $s_{\text{cm}} = \mathbf{-6.2595}$
• Beta Voice Clone Attack (EVAL-B-TSPOOF-SYN): $s_{\text{cm}} = \mathbf{-6.0606}$ (INVERTED: Higher than genuine)
• Beta Physical Replay Attack (EVAL-B-TSPOOF-REP): $s_{\text{cm}} = \mathbf{-5.8459}$ (INVERTED: Higher than genuine)

Mathematical Consequence:
Because $s_{\text{cm}}(\text{genuine}) < s_{\text{cm}}(\text{clone}) < s_{\text{cm}}(\text{replay})$, **no single 1-dimensional countermeasure threshold (τ_{cm}) can accept authentic Beta while rejecting both attacks.**
• Any threshold $\tau_{\text{cm}} \ge -6.06$ successfully rejects both attacks (4/4 attacks blocked on EVAL), but inevitably rejects authentic Beta ($1/3$ genuine acceptance).
• Any threshold $\tau_{\text{cm}} \le -6.26$ accepts authentic Beta, but admits both attacks.

Scientific Attribution Caveat:
This ordering mathematically proves the limitation of 1D linear cutoffs on this partition. It does NOT, by itself, establish that laptop microphone acoustics caused the inversion; determining the root cause (channel mismatch, speaker-specific vocal traits, or model training bias) requires controlled acoustic chamber recordings and further analysis.

3.2 Comparative Gate Evaluation Metrics (EVAL Partition: 21 Trials)

Decision Gate	Thresholds	TAR (%)	Human Imp Rej	Target Spoof Rej	All Spoof Rej
Standalone ASV	$\tau_{\text{asv}} = 0.3130$	66.7%	100.0%	50.0% (Failed)	80.0%
Standalone CM	$\tau_{\text{cm}} = -4.0694$	66.7%	25.0% (Failed)	75.0%	40.0%
Fused (Balanced)	$\tau_{\text{fus}} = 0.0783$	66.7%	100.0%	25.0% (Failed)	70.0%
Fused (Strict Pure)	$\tau_{\text{fus}} = 0.5052$	66.7%	100.0%	50.0% (Failed)	80.0%
Fused (Strict Gated)	$\tau_{\text{fus}} = 0.5052$, $\tau_{\text{cm}} = -4.0694$	33.3%	**100.0%**	**100.0% (4/4 Blocked)**	**100.0%**

3.3 Dataset Partition Roles & Experimental Discipline

To preserve scientific integrity, the project strictly partitions data into three formal roles:
1. **DEV Partition (21 Trials):** Sole training set for PyTorch logistic regression score fusion and policy threshold calibration.
2. **EVAL Partition (21 Trials):** Pilot validation feedback set (DEV-2). Because its failure modes directly motivated the architectural addition of the mandatory CM safeguard gate, EVAL is not treated as an untouched final test partition.
3. **TEST Partition (RESERVED):** An untouched blind evaluation partition to be collected and evaluated only after models, weights, and thresholds are permanently frozen.

PART 4: SYSTEM ARCHITECTURE & ENGINEERING IMPLEMENTATION

4.1 System Directory Layout & Codebase Structure

```
Integrated Spoof-Aware Speaker Verification/
■■■ src/
■ ■■■ api/server.py # FastAPI backend (strict_security, rate-limiting, CORS)
■ ■■■ core/
■ ■ ■■■ authentication_pipeline.py # End-to-end 5-stage pipeline with single outer try/finally
■ ■ ■■■ audio_processor.py # 16kHz resampling, RMS energy diagnostics, peak clipping
■ ■ ■■■ enrollment_manager.py # Multi-clip template enrollment with mandatory CM check
■ ■ ■■■ challenge_manager.py # Whisper ASR dynamic challenge, fuzzy match, 20s TTL
■ ■ ■■■ decision_engine.py # Multi-tier SASV decision logic & safeguard gate
■ ■■■ models/
■ ■■■ ecapa.py # SpeechBrain ECAPA-TDNN 192-d embedding extractor
■ ■■■ aasist.py # Raw-waveform AASIST GAT anti-spoof model wrapper
■■■ scripts/
■ ■■■ interactive_test.py # Live microphone enrollment & testing CLI tool
■ ■■■ calibrate_live_pipeline.py # Reproducible DEV-only threshold selection script
■ ■■■ test_backend.py # 7-stage automated backend integration test suite
■■■ data/
■ ■■■ voices/ # Structured permanent voice archive
■ ■ ■■■ enrolled/{username}/ # Archived enrollment WAV recordings
■ ■ ■■■ testing/{username}/ # Archived test probes tagged with ACCESS_GRANTED / DENIED
■ ■■■ enrolled_users/ # Persisted NPY voiceprints and JSON metadata
■ ■■■ benchmark/ # DEV/EVAL manifests, audio, and SASV 2022 evaluation data
■■■ models/live_pipeline_metadata.json # Calibrated weights, operating policies, and metadata
```

4.2 Production Security Engineering Principles Implemented

• **Strict Server-Controlled Policy:** The server enforces SERVER_POLICY = 'strict_security'. Clients cannot pass form fields to downgrade security.
• **Mandatory Enrollment Anti-Spoof Verification:** The /api/v1/enroll endpoint strictly evaluates AASIST CM scores on uploaded clips, rejecting synthetic audio with HTTP 400.
• **Zero Cross-Request State Pollution:** The entire authentication execution is wrapped in a single outer try...finally block, guaranteeing that early exits (silent audio, missing tokens, unverified accounts) unconditionally restore the baseline engine state.
• **Permanent Structured Archiving:** Microphone recordings are automatically organized into data/voices/enrolled/{username}/ and data/voices/testing/{username}/ with ISO timestamps and verdict tags.

4.3 How to Run & Validate the System

The project includes multiple execution workflows for testing and demonstration:

```
1. Run Automated Backend Integration Test Suite:
PS> python scripts/test_backend.py
(Executes 7 test suites: enrollment spoof gate, challenge replay, pipeline fusion,
tandem decision, mandatory safeguard gate, FastAPI endpoints, and enrollment API security. Code 0).

2. Run Reproducible DEV-Only Policy Calibration:
PS> python scripts/calibrate_live_pipeline.py
(Trains PyTorch fusion layer on DEV, calculates exact arithmetic thresholds, scores EVAL, updates
metadata).

3. Start FastAPI REST Server:
PS> python src/api/server.py
(Listens on http://127.0.0.1:8000; exposes OpenAPI docs at http://127.0.0.1:8000/docs).

4. Run Interactive Microphone Enrollment & Authentication Tool:
PS> python scripts/interactive_test.py
(Option 1: Enroll Voice | Option 2: Live Authentication | Option 3: List Users | Option 4: View
Archived Voices).
```

PART 5: TECHNICAL ROUND INTERVIEW PLAYBOOK

5.1 The 60-Second Elevator Pitch

60-Second Pitch for Interviewers

"I built an Integrated Spoof-Aware Speaker Verification (SASV) system designed to secure voice authentication against generative AI deepfakes and physical replay attacks.

Traditional voice biometrics have a fatal blindspot: if an attacker clones a victim's voice using modern neural vocoders, speaker verification models accept them because the voice similarity is exceptionally high. My system integrates three synchronized neural architectures: SpeechBrain's ECAPA-TDNN to verify speaker identity via 192-dimensional embeddings, AASIST with spectro-temporal graph attention networks operating directly on raw waveforms to detect synthetic artifacts, and OpenAI Whisper to enforce a single-use dynamic spoken challenge with a 20-second TTL to defeat replay attacks.

Crucially, I discovered that pure linear score fusion still leaks voice clones because high voice similarity mathematically overwhelms negative countermeasure scores. To solve this, I designed a non-linear tandem safeguard gate that mandates passing the countermeasure floor regardless of voice similarity, achieving 100% target attack rejection on our benchmark evaluation set. The entire backend is built with FastAPI, CUDA GPU acceleration, strict request isolation, and persistent audio archiving."

5.2 Deep-Dive Question & Answer Bank for Technical Rounds

Q1: Why can't traditional speaker verification (ASV) models detect voice clones?

Traditional ASV models (like ECAPA-TDNN or x-vectors) are trained with angular margin loss (AAM-Softmax) solely to maximize inter-speaker distance and minimize intra-speaker distance. They are mathematically optimized to find identity traits. Modern voice clones capture speaker formants and vocal tract traits so accurately that the clone's embedding is closer to the true user than a real recording taken on a different microphone. ASV models have no architectural mechanism to evaluate acoustic naturalness.

Q2: How does AASIST operate on raw audio, and why not use Mel-spectrograms?

Spectrograms discard raw phase information by taking the squared magnitude of the STFT. However, neural vocoders and physical loudspeakers introduce subtle temporal phase incoherencies and high-frequency dispersion that are visible only in the raw waveform. AASIST uses a SincNet filterbank to learn bandpass filters directly from 16 kHz raw waveforms, followed by heterogeneous graph attention networks that pass attention messages between temporal and spectral nodes.

Q3: Why did pure linear score fusion fail against Beta voice clones?

In linear fusion ($s_{\text{fus}} = w_{\text{asv}} s_{\text{asv}} + w_{\text{cm}} s_{\text{cm}} + b$), the trained ASV weight was +12.72 while the CM weight was +0.17. For Beta voice clones, the voice similarity was so high ($s_{\text{asv}} = +0.7181$) that $(12.72 \times 0.7181 = +9.13)$ completely dominated the countermeasure warning ($0.17 \times -6.06 = -1.07$), yielding $s_{\text{fus}} = +4.5899$. Under pure linear fusion, the attack was accepted. This proved that linear combinations cannot represent the logical requirement: 'Identity AND Authenticity'.

Q4: What is the Beta CM score inversion phenomenon?

On our evaluation partition, authentic speaker Beta scored $s_{\text{cm}} = -6.2595$, while Beta's synthetic clone scored -6.0606 and replay attack scored -5.8459. Because the genuine score is lower than both attack scores, no 1-dimensional cutoff can accept genuine Beta while rejecting both attacks. This proves that countermeasure models experience domain gaps and that strict security incurs a measured trade-off in genuine rejection (TAR dropped to 33.3%).

Q5: How did you ensure your threshold calibration was scientifically valid?

I implemented a reproducible, DEV-only threshold selection procedure. Thresholds are derived strictly from the development set using exact arithmetic formulas (e.g. Balanced threshold $= (\max S_{\text{imp}} + \min S_{\text{gen}}) / 2 = +0.0783$). The EVAL partition was strictly quarantined as a validation feedback set to diagnose failure modes, and a clean TEST partition is reserved for untouched evaluation once models are frozen.

Q6: How did you guarantee cross-request state isolation in FastAPI?

In a multi-user API, modifying pipeline settings on a shared singleton can leak weakened policies across requests. I wrapped the entire 5-stage authentication pipeline inside a single outer `try...finally` block. The baseline engine state is snapshotted before any processing, and the `finally` block unconditionally restores the original policy, thresholds, and CM gate on every return path—including early quality denials, challenge failures, and unhandled exceptions.